Image in Frequency Domain

Deep Learning for Computer Vision

Image in Frequency Domain

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Non-Linear Filters

Different types of noise in images



Reducing Salt-and-Pepper noise using Gaussian filters



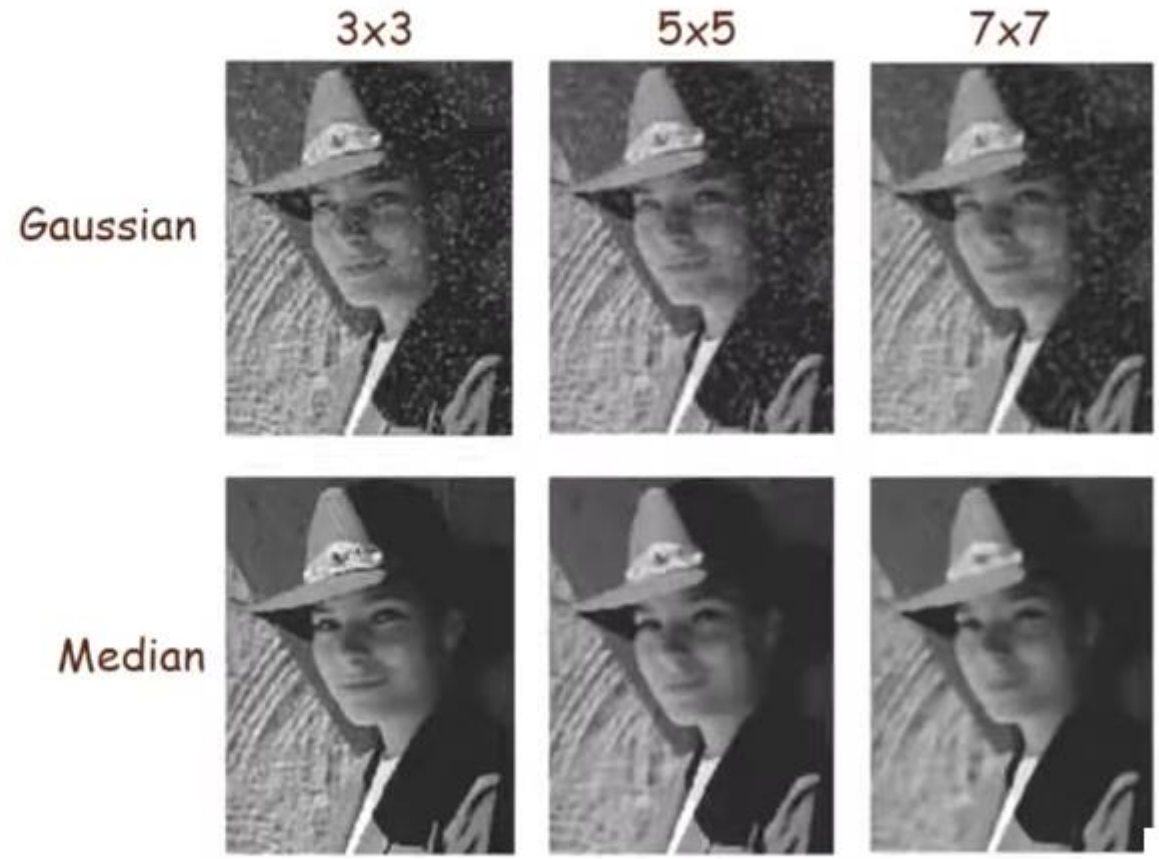
See the problem? What do we do?

Credit: S Seitz, Univ of Washington & J Košecká, George Mason University

Non-Linear Filters: Median Filter

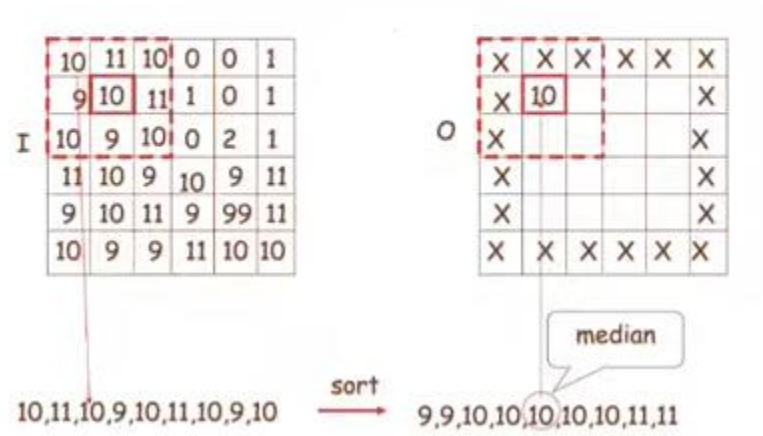
- Replace each pixel with MEDIAN value of all pixels in neighbourhood

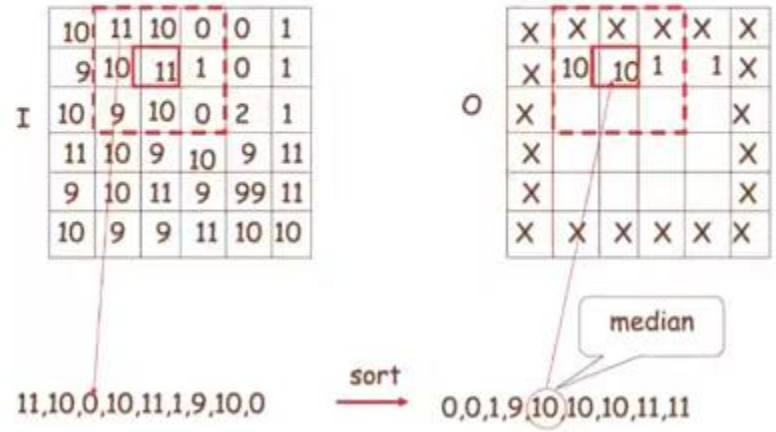
- Properties:
  - Non-linear
  - Does not spread noise
  - Can remove spike noise
  - Robust to outliers, but not good for Gaussian noise

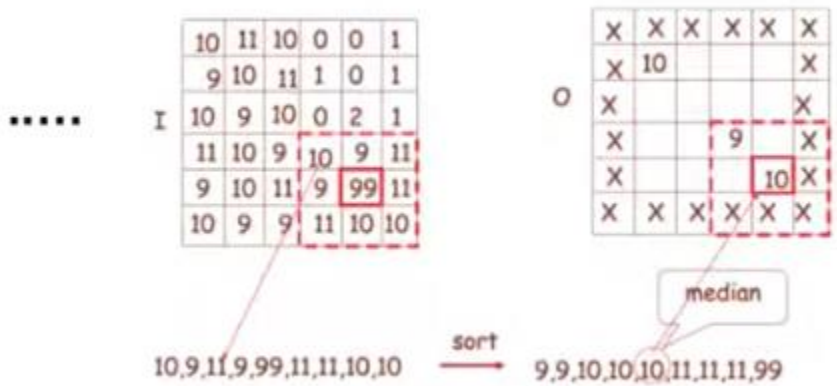


Credit: J Košecká, George Mason University

Median Filter: Example







Notice how the outlier pixel value (99) got filtered out

Credit: J Košecká, George Mason University

Non-Linear Filters: Bilateral Filtering

- Noise removal comes at expense of image blurring at edges
- **Bilateral filtering:** Simple, non-linear edge-preserving smoothing
- Reject (in a soft manner) pixels whose values differ too much from the central pixel value.
- Output pixel value is weighted combination of neighboring pixel values: $g(i,j) = \frac{\sum_{k,l} I(k,l) w(i,j,k,l)}{\sum_{k,l} w(i,j,k,l)}$
- Data-dependent bilateral weight function composed of domain and range kernel:

$$w(i,j,k,l) = \exp\left(-\frac{(i-k)^2 + (j-l)^2}{2\sigma_d^2} - \frac{||I(i,j) - I(k,l)||^2}{2\sigma_r^2}\right)$$

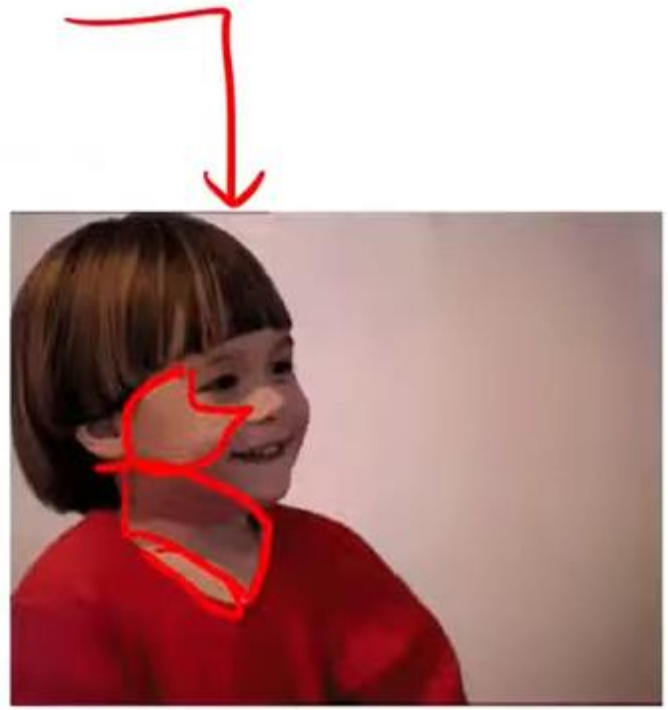


Credit: Wikipedia; CVOnline

Non-Linear Filters: Bilateral Filtering

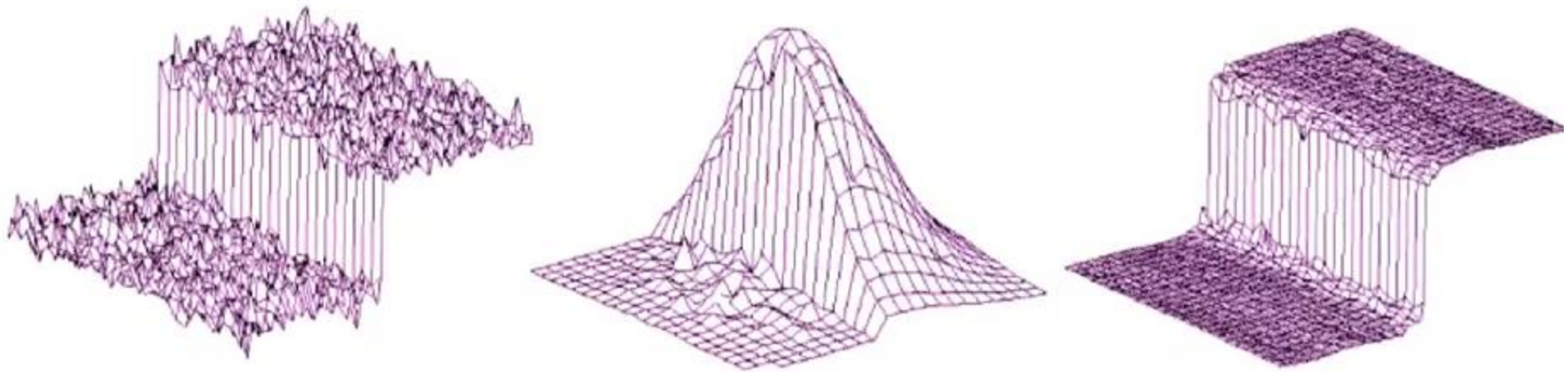
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Credit: Wikipedia; CVOnline

Bilateral Filters: Example



Given region in an image

23 x 23 filter centered
two pixels to right of step
in previous figure

Filtered output

Credit: *CVOnline*

What do we lose in a low-resolution image?



Credit: Derek Hoeim, UIUC; James Hays, Gatech

Fourier

Jean Baptist Joseph Fourier (1768-1830) had an idea in (1807).

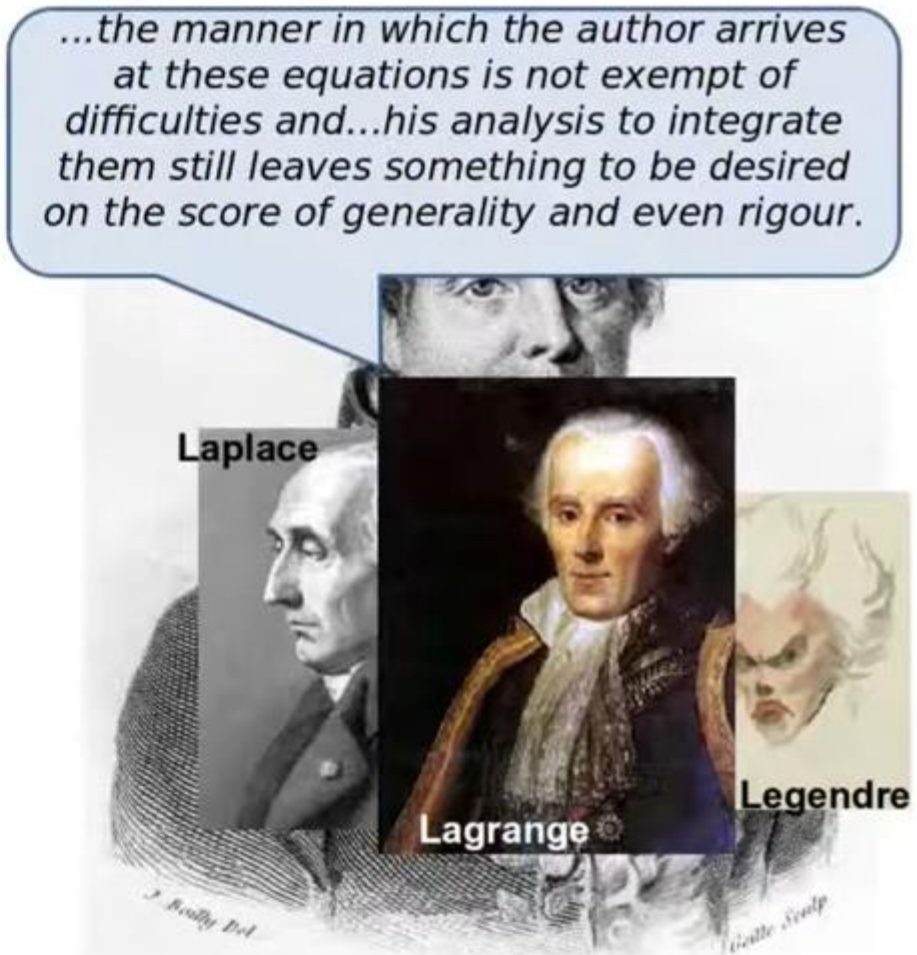
- **Idea:** Any univariate function can be rewritten as a weighted sum of sines and cosines of different frequencies.

Fourier

Jean Baptist Joseph Fourier (1768-1830) had an idea in (1807).

- **Idea:** Any univariate function can be rewritten as a weighted sum of sines and cosines of different frequencies.
- **Of course, what's new?**
  - Many including Lagrange, Laplace, Poisson and other big wigs did not believe him
  - Not translated into English until 1878!
- **(Mostly) true!**
  - Called Fourier Series
  - Some subtle restrictions

Credit: James Hays

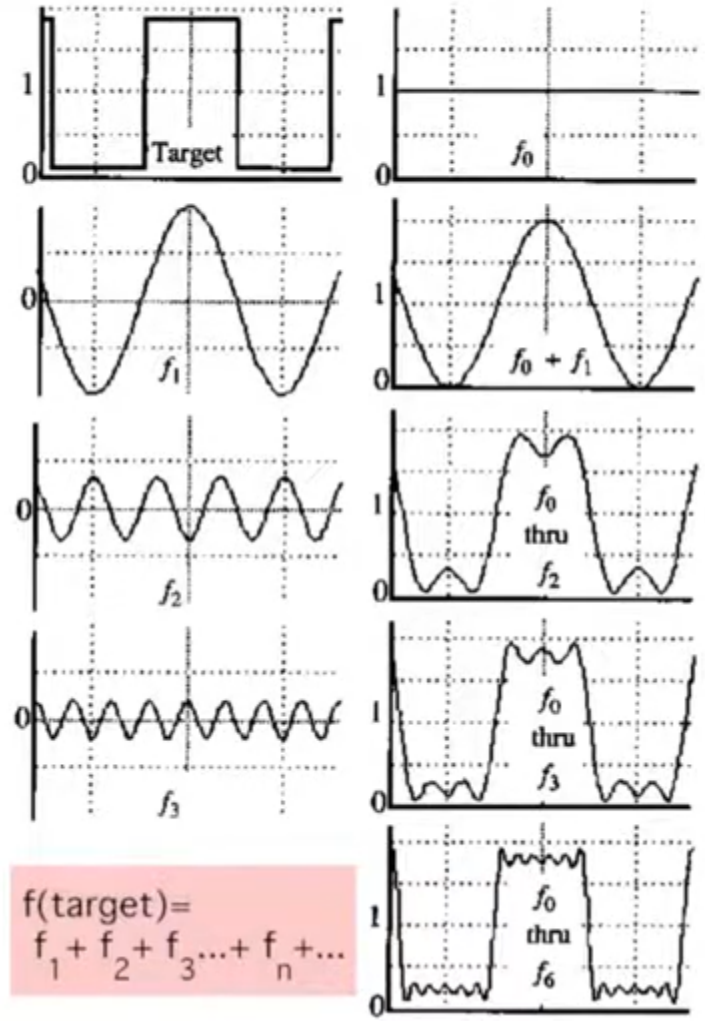


A sum of sines

- Building block:

$$A \sin(\omega x + \phi)$$

- Add enough of them to get any signal f(x) you want!



Credit: James Hays, Gatech

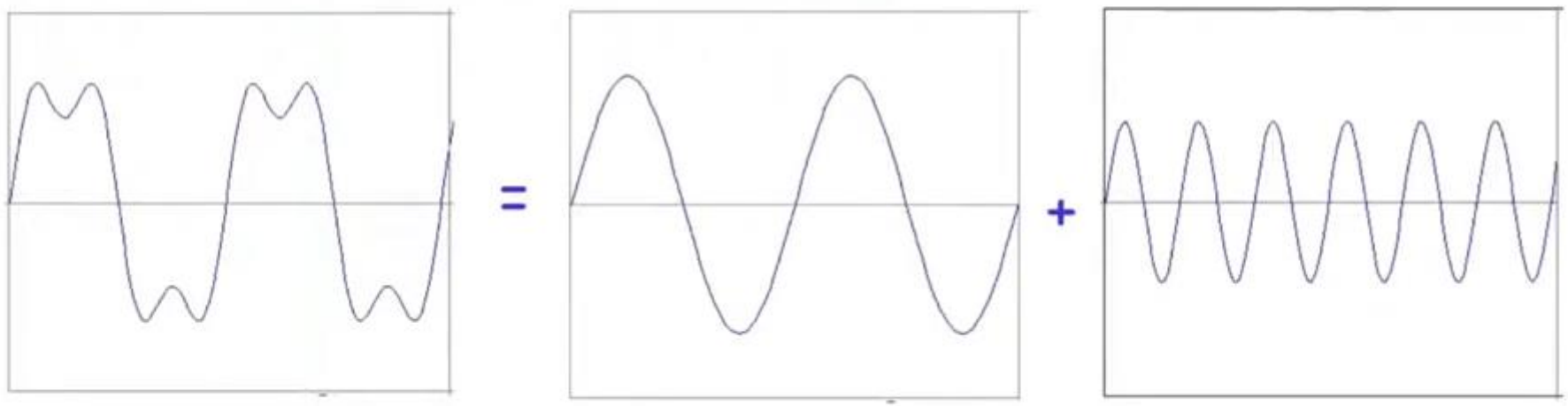
The Fourier Spectrum

Example: $g(t) = \sin(2pft) + (1/3)\sin(2p(3f)t)$



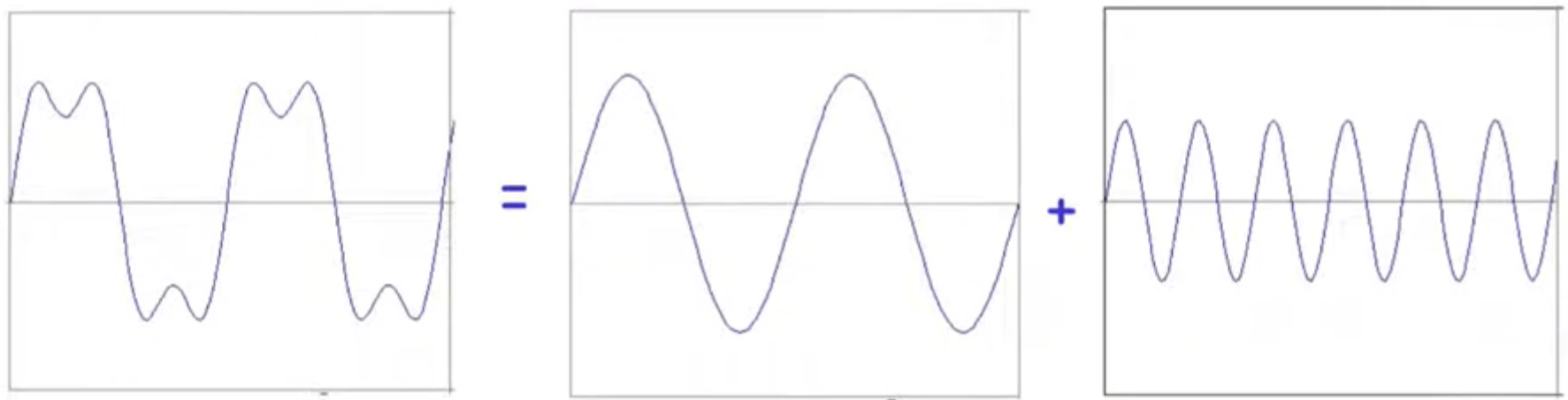
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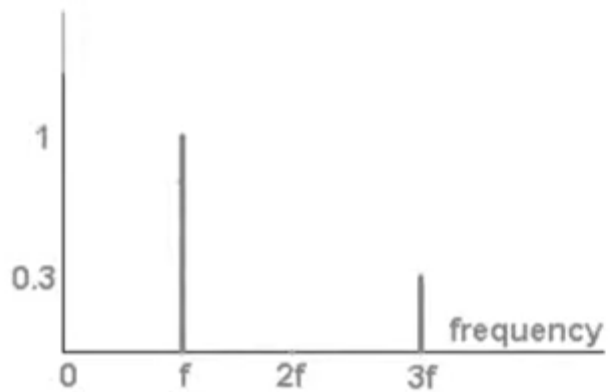
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The Fourier Spectrum

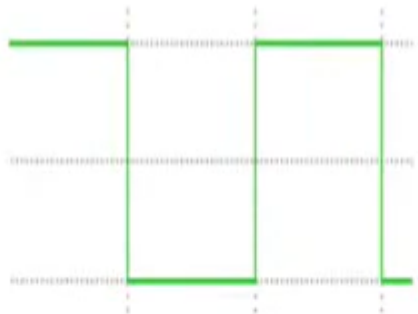
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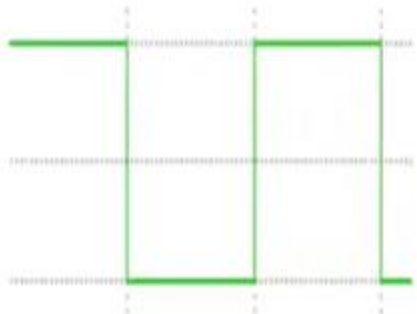


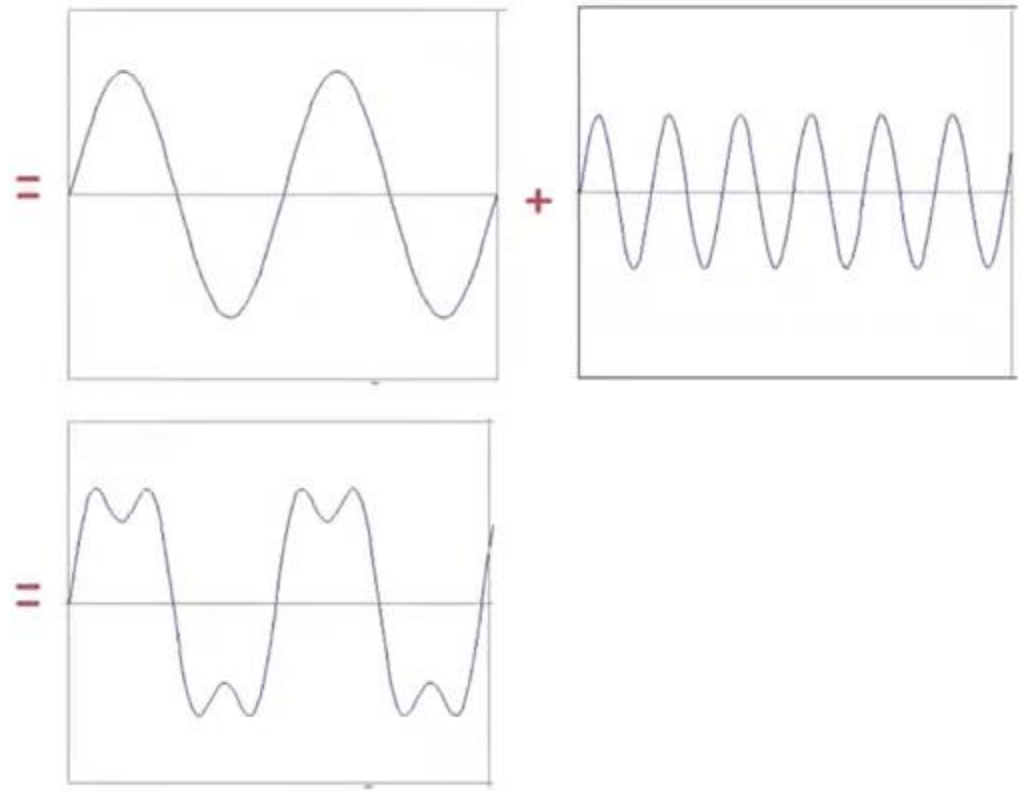
Credit: Alexei Efros, UCBerkeley; James Hays, Gatech

The Fourier Spectrum



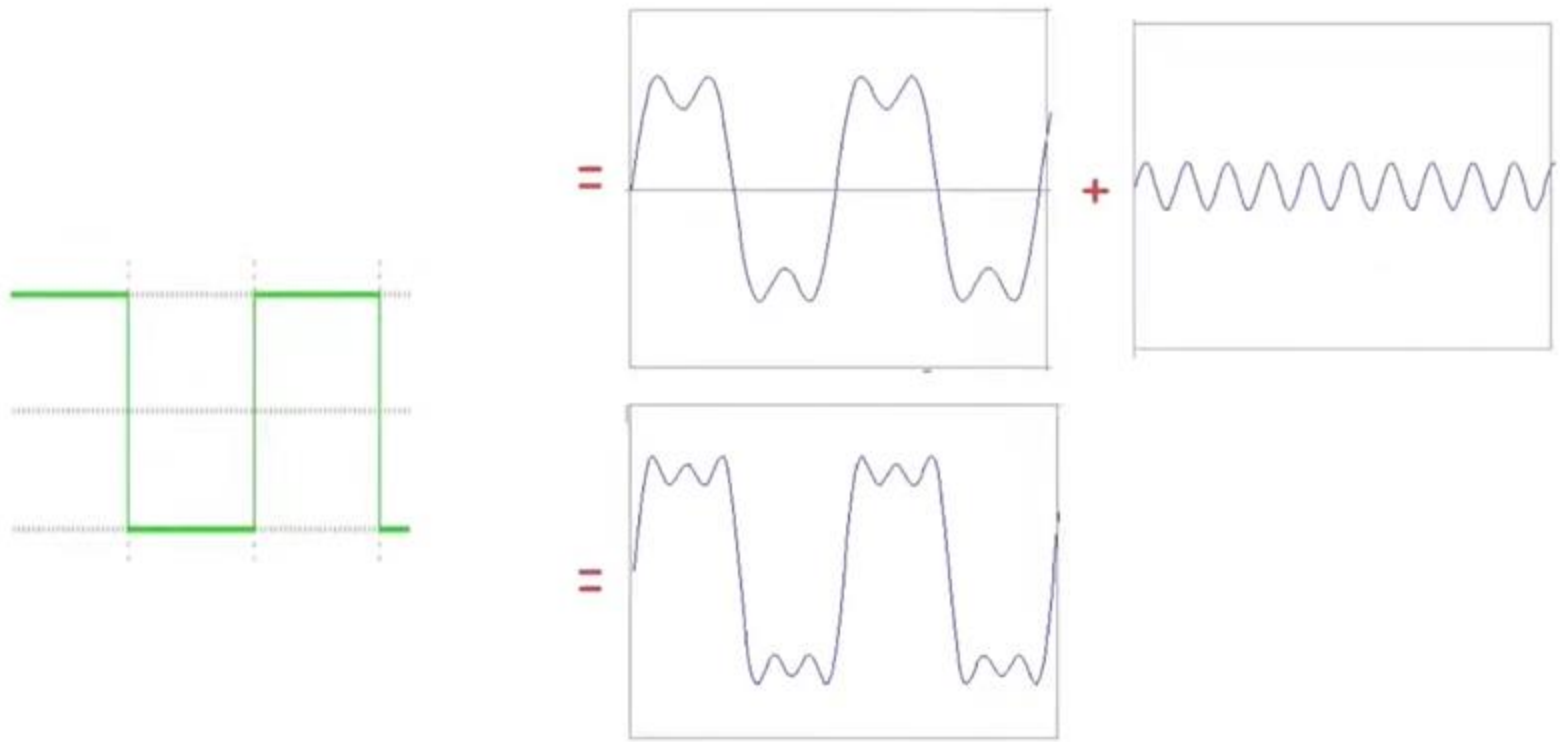
The Fourier Spectrum





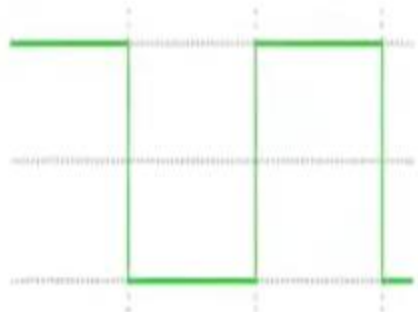
Credit: James Hays, Gatech

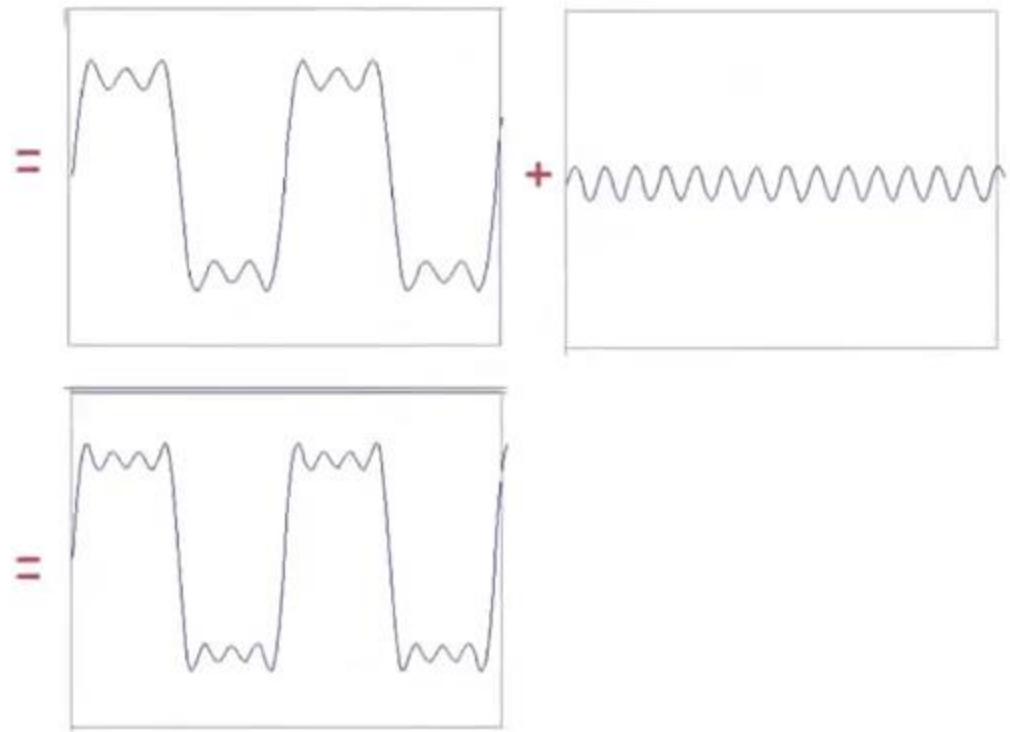
The Fourier Spectrum



Credit: James Hays, Gatech

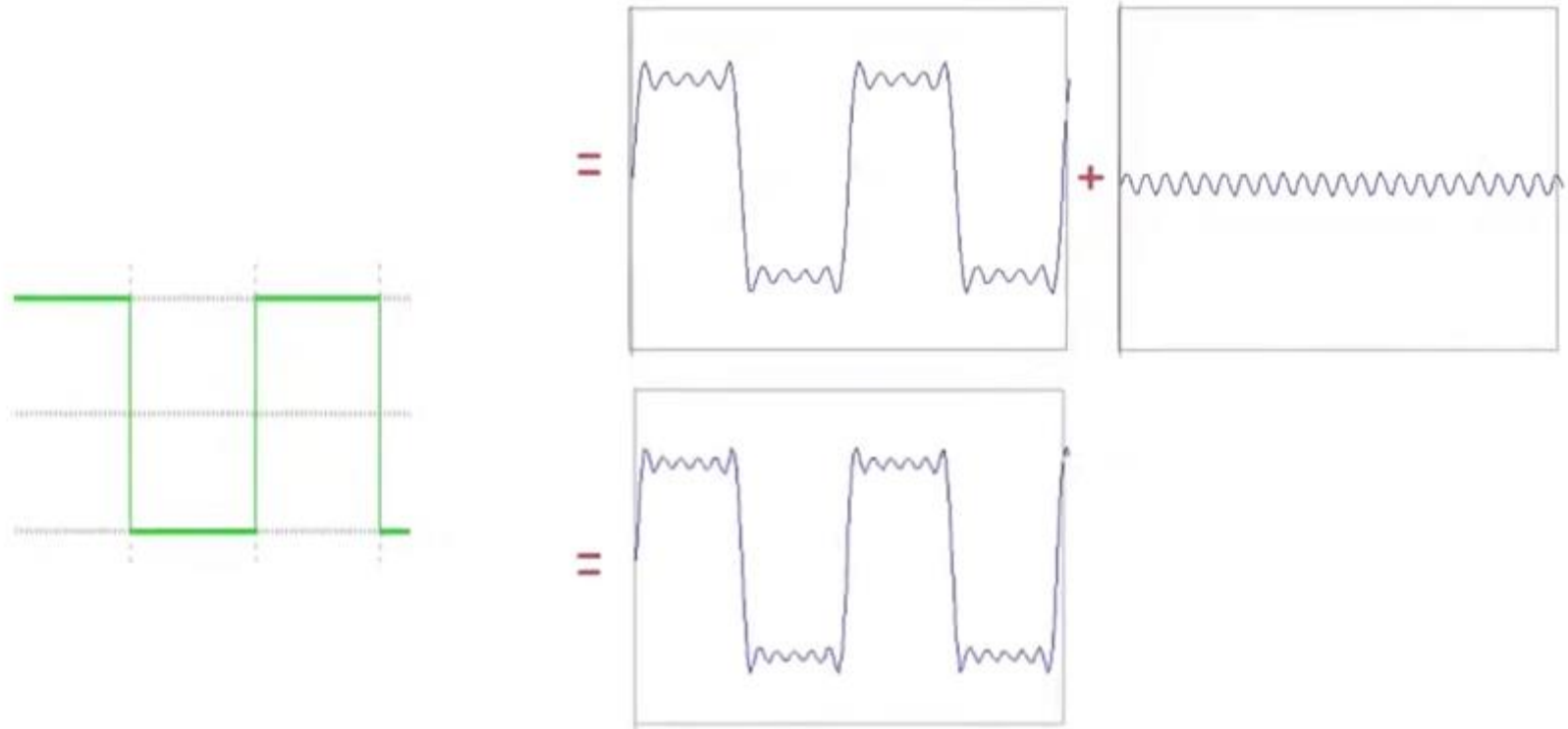
The Fourier Spectrum





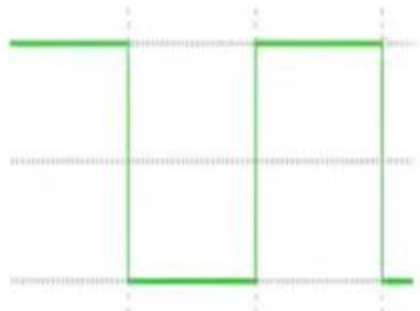
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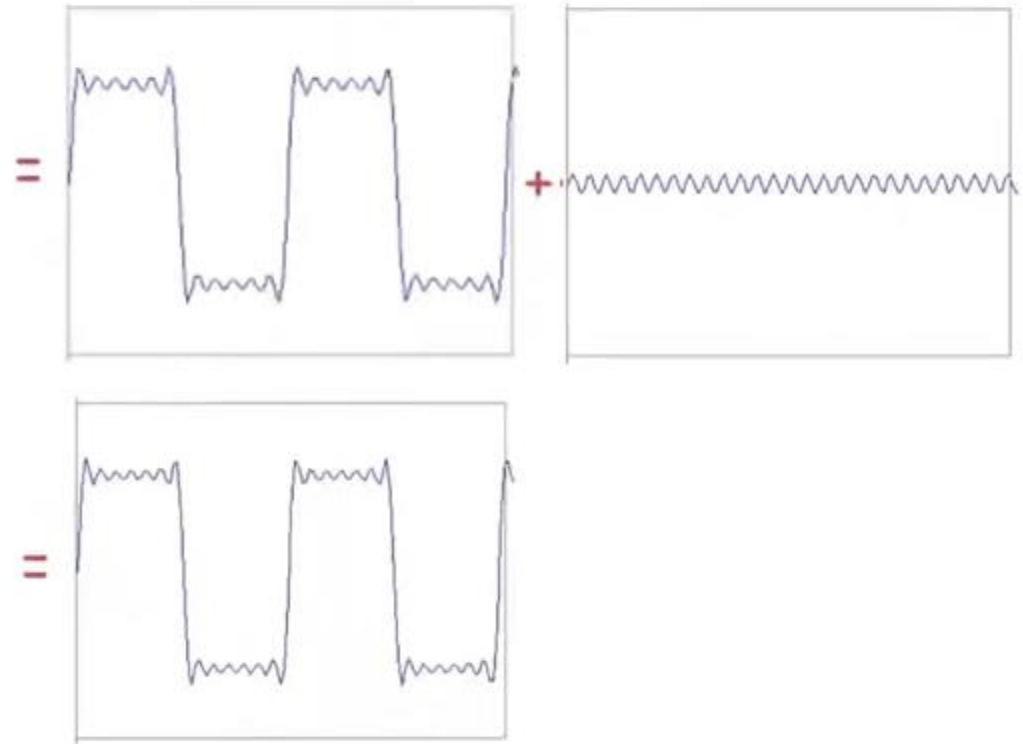
The Fourier Spectrum



Credit: James Hays, Gatech

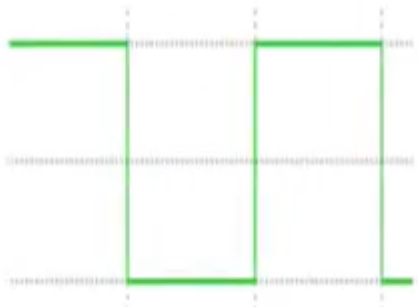
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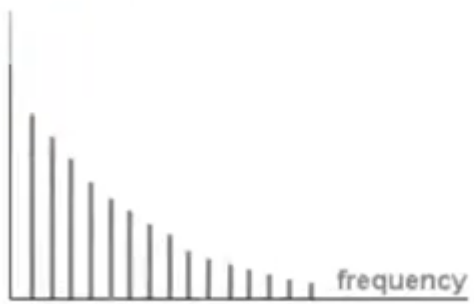


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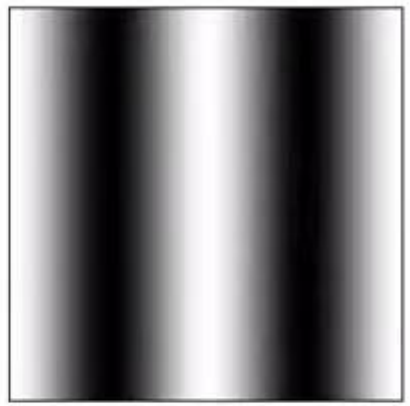
$$= A \sum_{1}^{\infty} \frac{1}{k} \sin(2\pi k t)$$

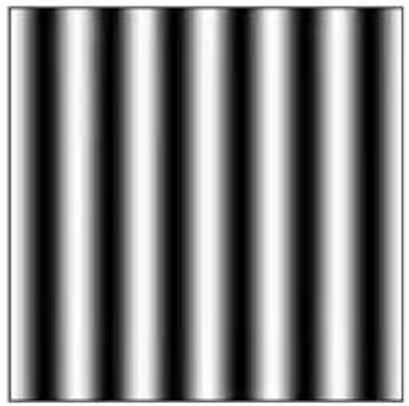


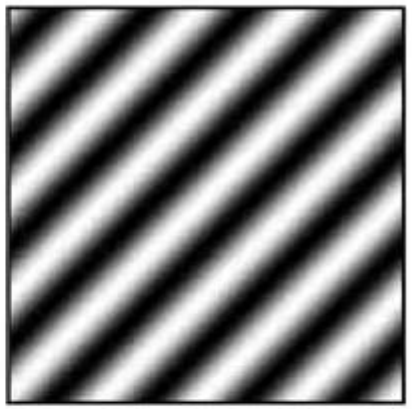
Credit: James Hays, Gatech

Fourier Analysis of Images

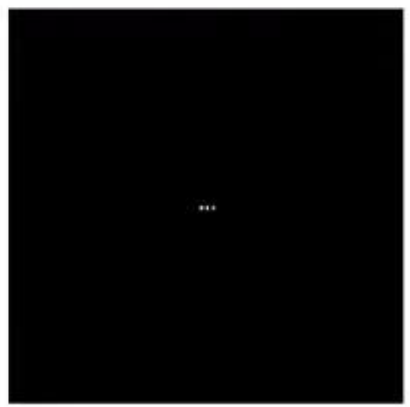
Intensity Image

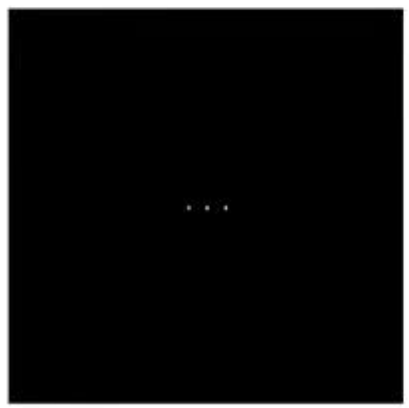






Fourier Image

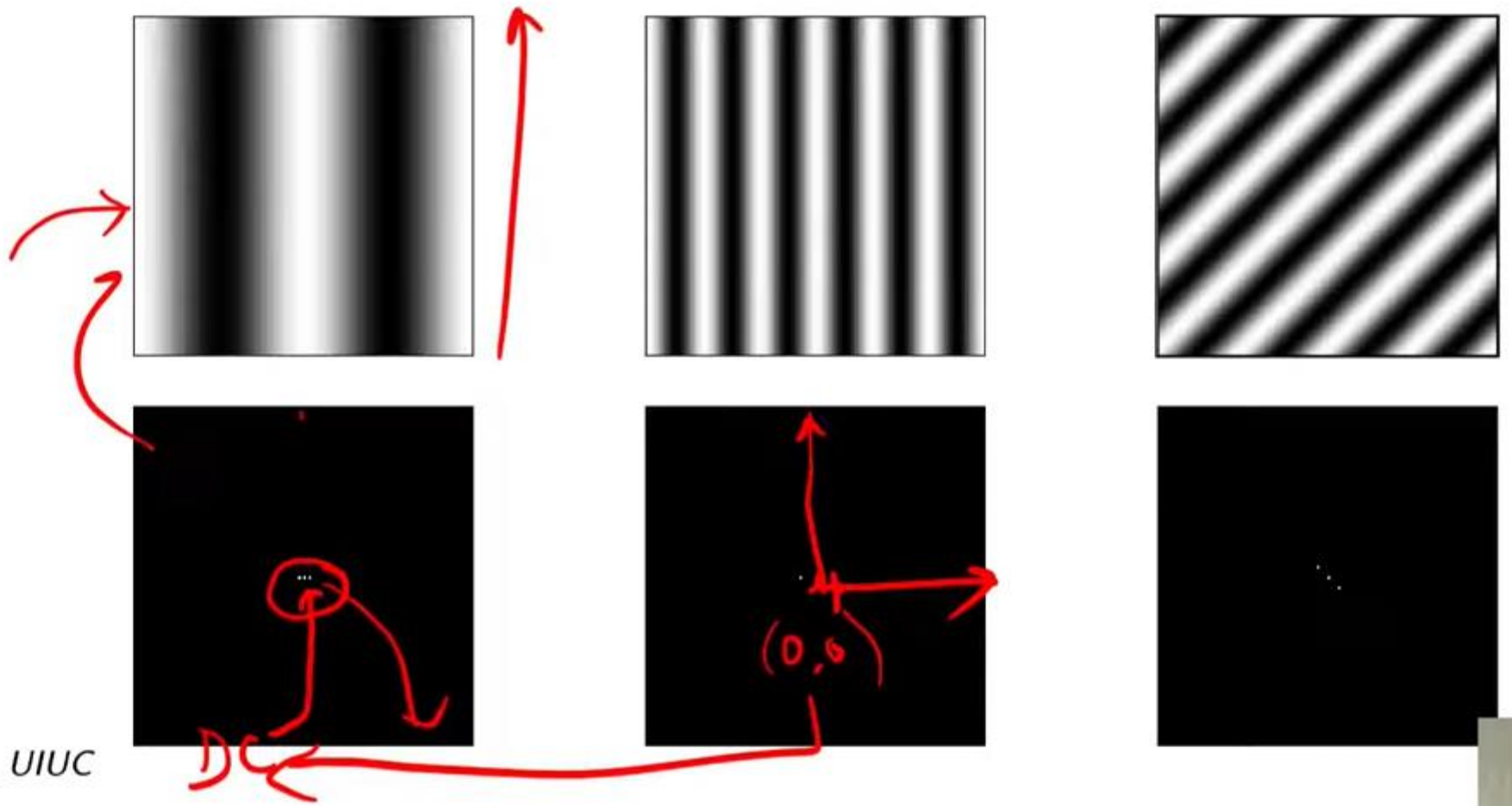






Credit: Derek Hoeim, UIUC

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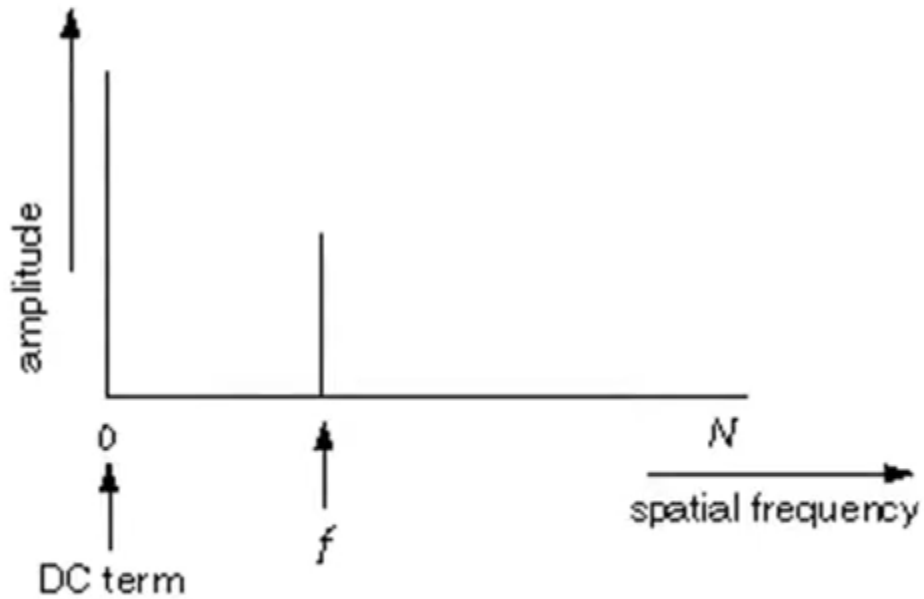


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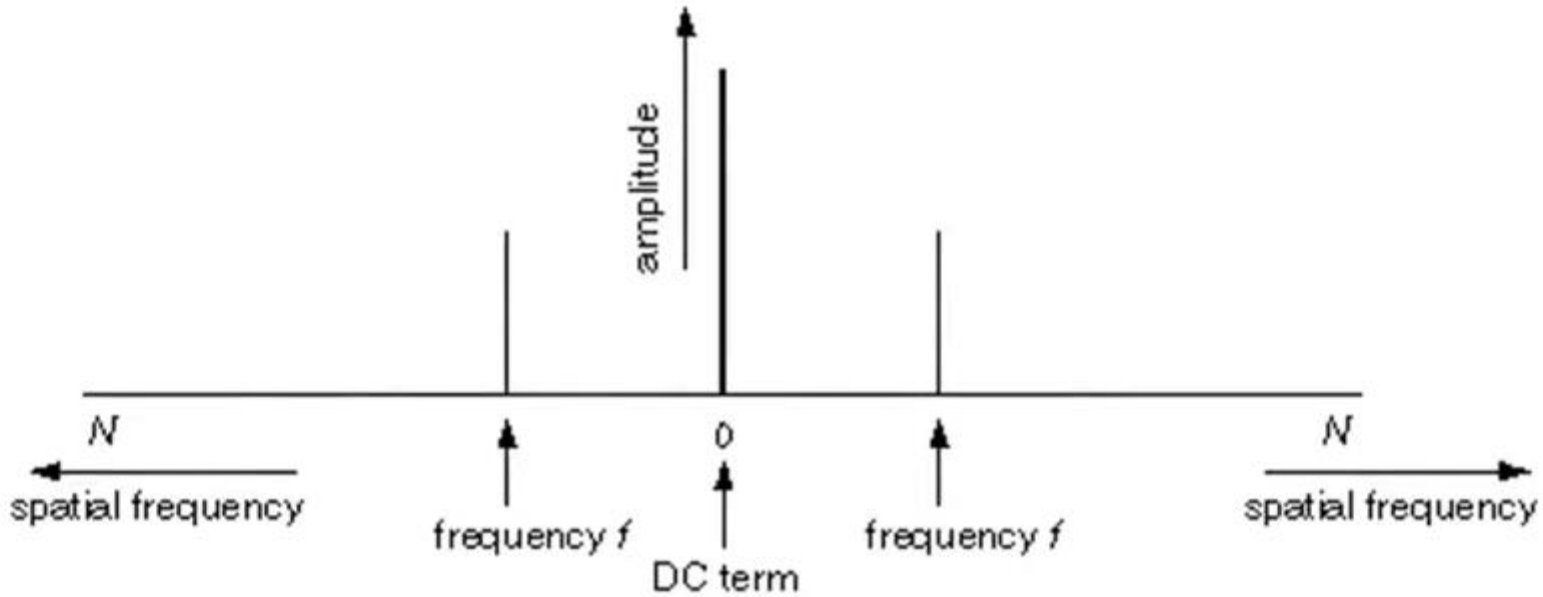
- Encodes a whole series of sinusoids through a range of spatial frequencies from zero all the way up to '*Nyquist frequency*' (more on this later)

- Signal containing only a single spatial frequency of frequency f is plotted as:
  - a single peak at point f along spatial frequency axis
  - height of that peak corresponding to the amplitude, or contrast of that sinusoidal signal



Fourier Analysis of Images

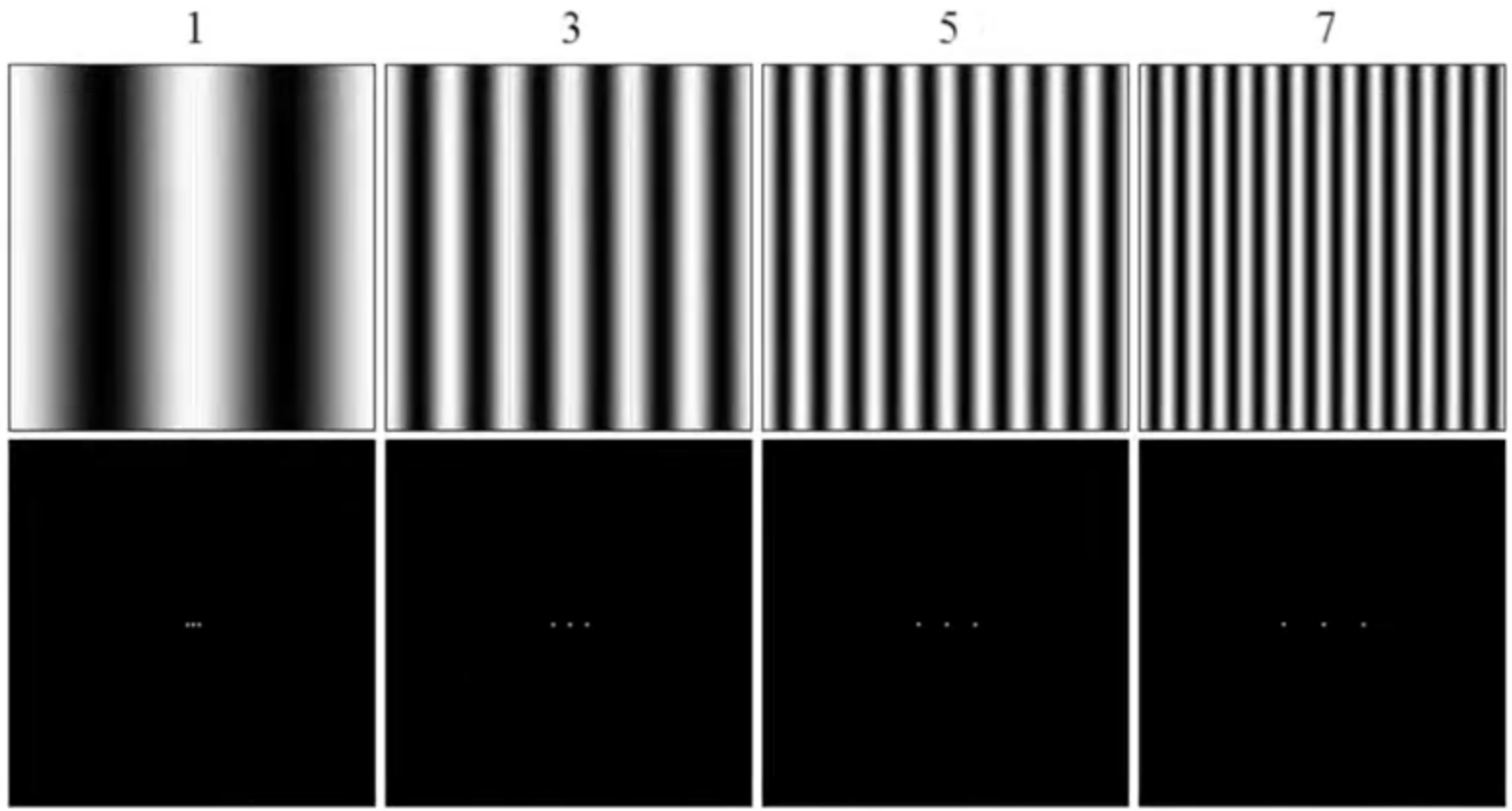
Mirror-image reflections along the axes



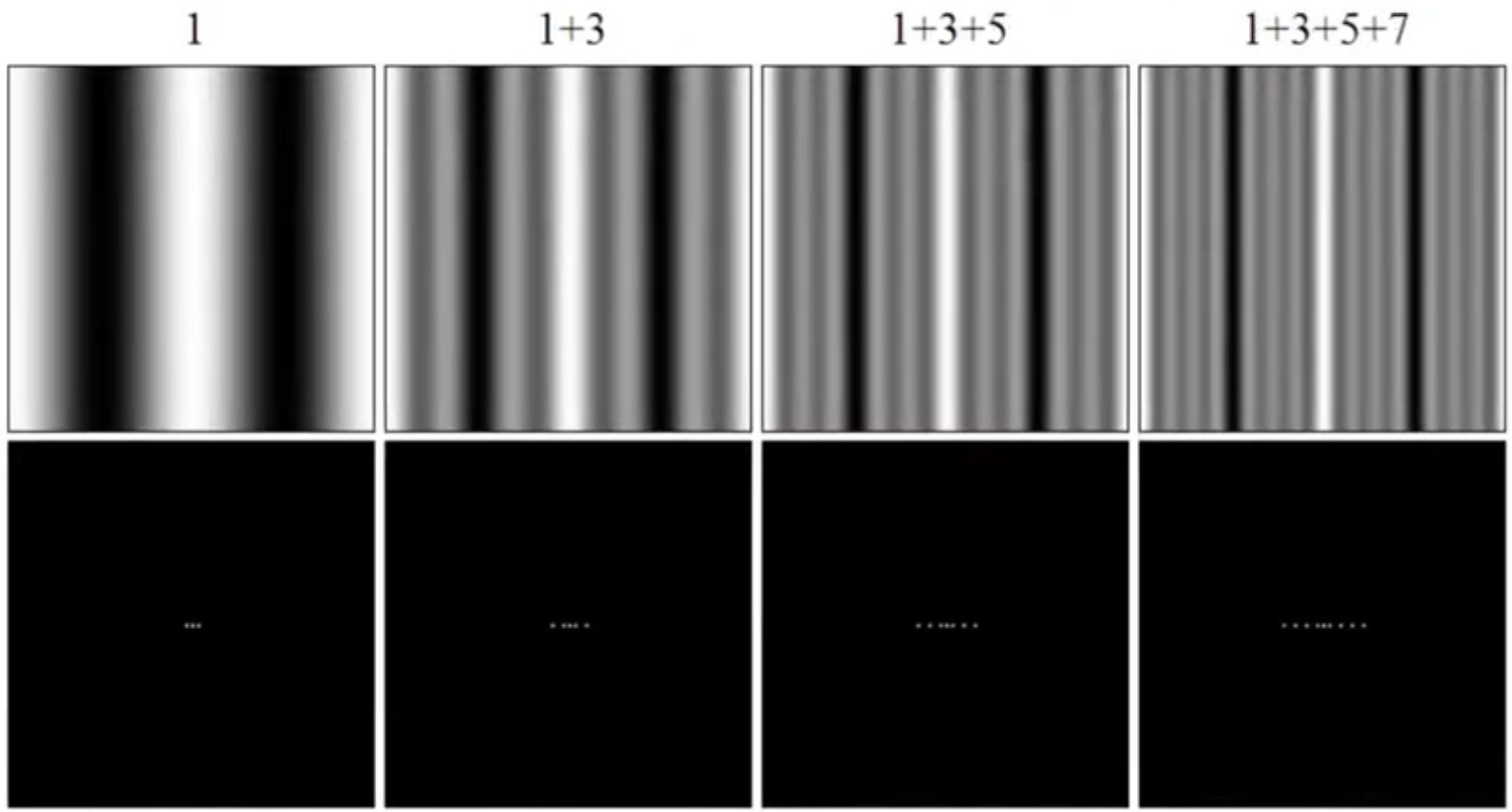
Why? See

http://dsp.stackexchange.com/questions/4825/why-is-the-fft-mirrored

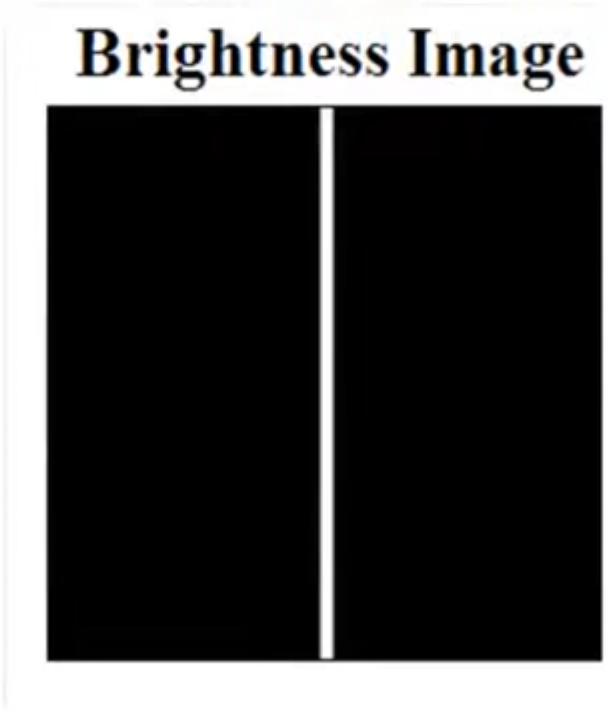
Fourier Analysis of Images: Examples

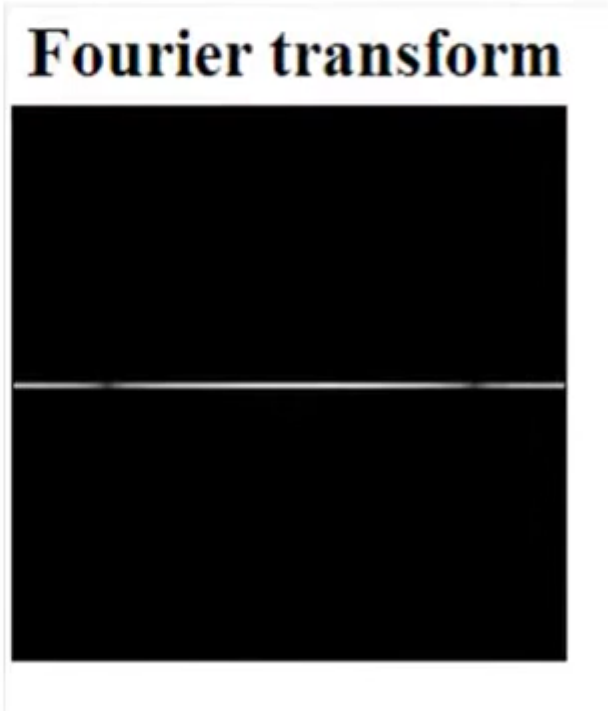


Fourier Analysis of Images: Examples

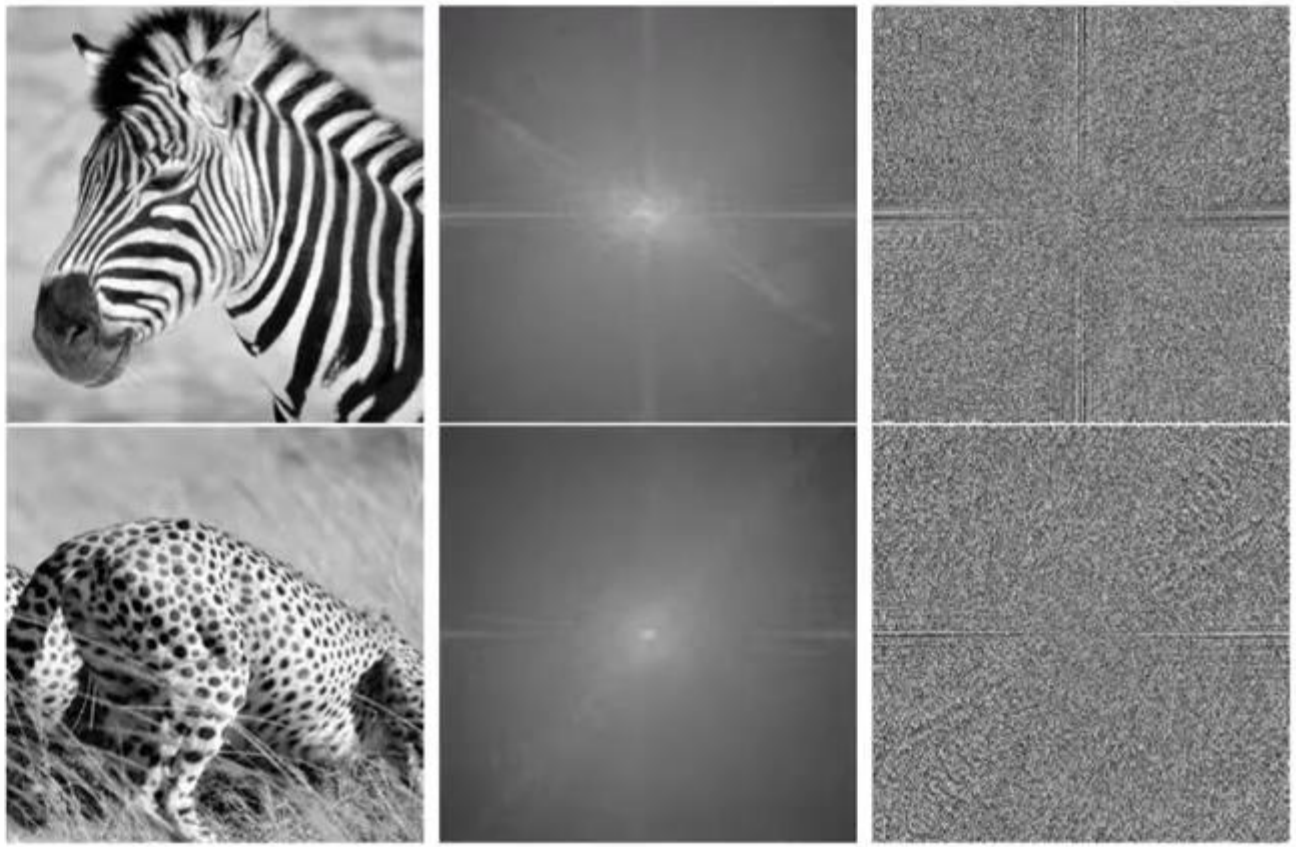


Fourier Analysis of Images: Examples





Fourier Analysis of Images: Examples



Credit: Forsyth and Ponce, Computer Vision: A Modern Approach, 2003

Fourier Transform: Magnitude and Phase

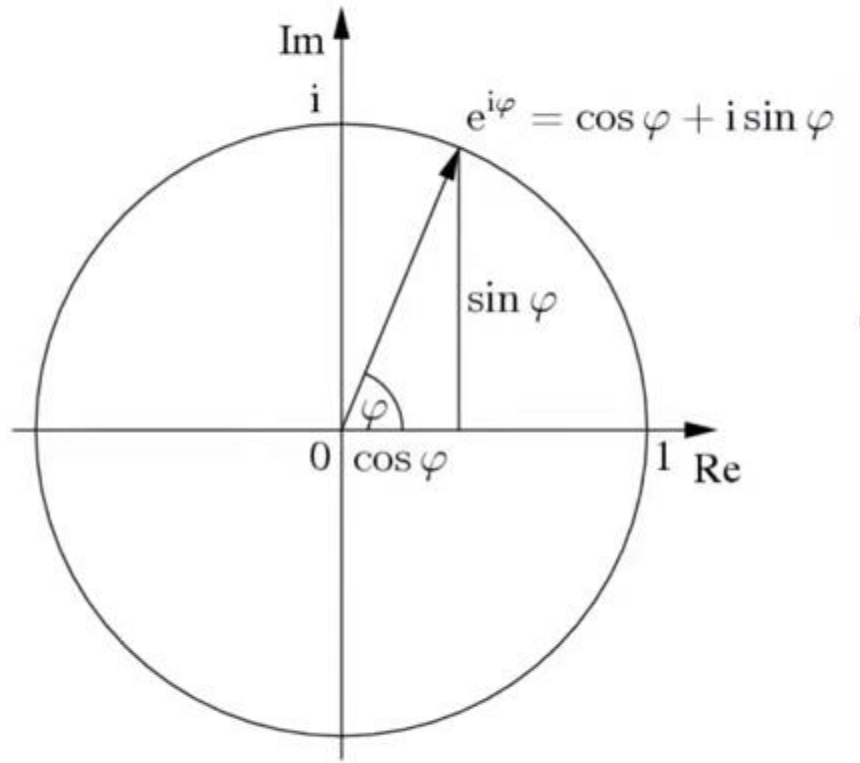
- Fourier transform stores the **magnitude** and **phase** at each frequency
  - For mathematical convenience, this is often denoted in terms of real and complex numbers
  - **Magnitude** encodes how much signal is there at a particular frequency

$$A = \pm\sqrt{Re(\varphi)^2 + Im(\varphi)^2}$$

  - **Phase** encodes spatial information (indirectly)

$$\phi = \tan^{-1}\frac{Im(\varphi)}{Re(\varphi)}$$

Credit: Derek Hoeim, UIUC



Credit: Wikimedia Commons

Continuous vs Discrete Fourier Transform

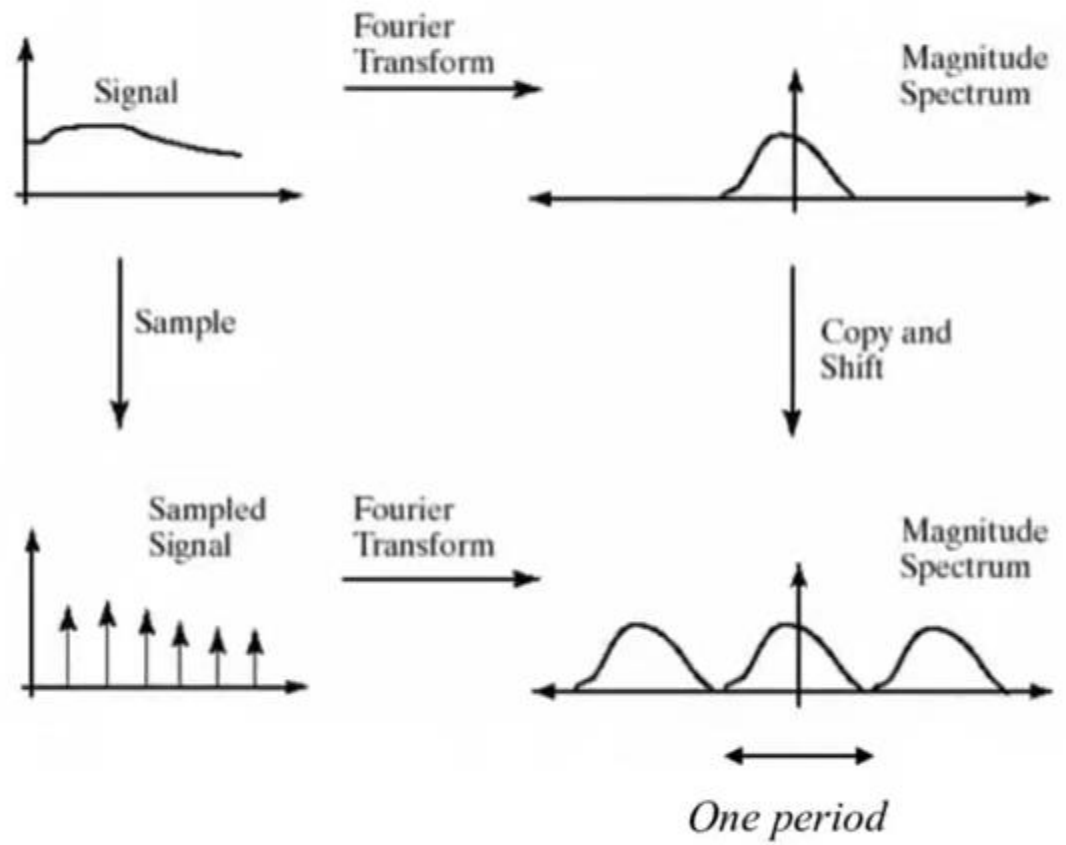
- **Continuous Fourier transform (FT):**

$$H(\omega) = \int_{-\infty}^{\infty} h(x) e^{-j\omega x} dx$$

- **Discrete Fourier Transform (DFT):**

$$H(\omega) = \sum_{0}^{N-1} h(x) e^{-j\frac{2\pi k x}{N}} dx$$

where N is the length of the sampled signal.



Credit: Elgammal, Rutgers University

Convolution Theorem

- Fourier transform of convolution of two functions is product of their Fourier transforms:

$$F[g * h] = F[g]F[h]$$

- **Convolution** in spatial domain can be obtained through **multiplication** in frequency domain!

$$g * h = F^{-1}[F[g]F[h]]$$

Credit: James Hays, Gatech

Properties of Fourier Transform

Property	Signal		Transform
superposition	$f_1(x) + f_2(x)$		$F_1(\omega) + F_2(\omega)$
shift	$f(x - x_0)$		$F(\omega)e^{-j\omega x_0}$
reversal	$f(-x)$		$F^*(\omega)$
convolution	$f(x) * h(x)$		$F(\omega)H(\omega)$
correlation	$f(x) \otimes h(x)$		$F(\omega)H^*(\omega)$
multiplication	$f(x)h(x)$		$F(\omega) * H(\omega)$
differentiation	$f'(x)$		$j\omega F(\omega)$
domain scaling	$f(ax)$		$1/aF(\omega/a)$
real images	$f(x) = f^*(x)$	$\Leftrightarrow$	$F(\omega) = F(-\omega)$
Parseval's Theorem	$\sum_x [f(x)]^2$	$=$	$\sum_\omega [F(\omega)]^2$

Credit: Szeliski, Computer Vision: Algorithms and Applications, 2010

Fast Fourier Transform

- Number of arithmetic operations to compute Fourier transform of N numbers (i.e., function defined at N points) is.... proportional to N^2
- Possible to reduce this to $N \log N$ using **Fast Fourier Transform (FFT)**
- FFT is a **recursive divide-and-conquer algorithm** for computing DFT

For more, see https://www.karlsims.com/fft.html

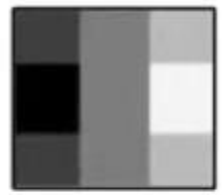
Filtering in Spatial Domain

1	0	-1
2	0	-2
1	0	-1

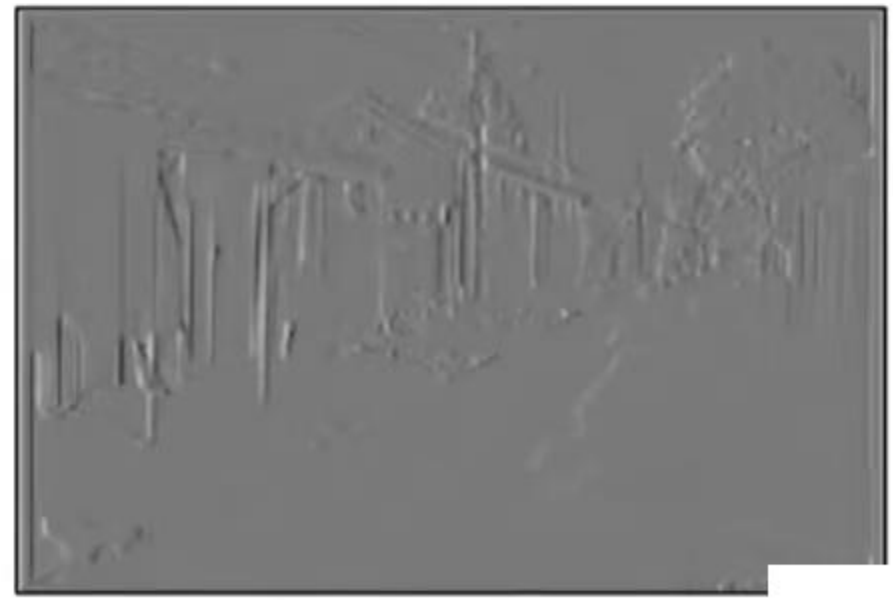
Intensity Image



*

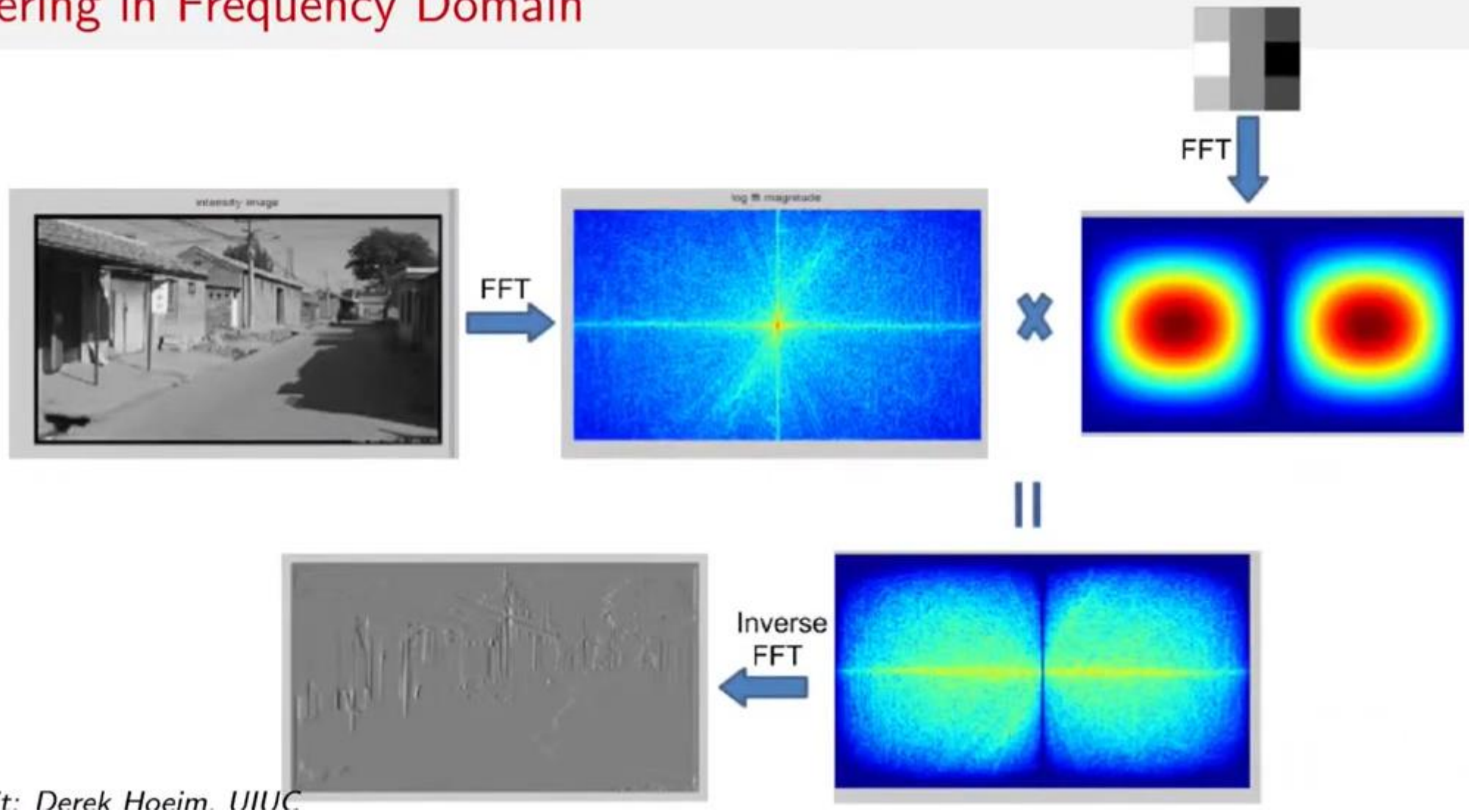


=



Credit: Derek Hoeim

Filtering in Frequency Domain



Credit: Derek Hoiem, UIUC

Low-Pass and High-Pass Filters

- **Low-Pass Filters:** Filters that allow low frequencies to pass through (block high frequencies). Example?

Low-Pass and High-Pass Filters

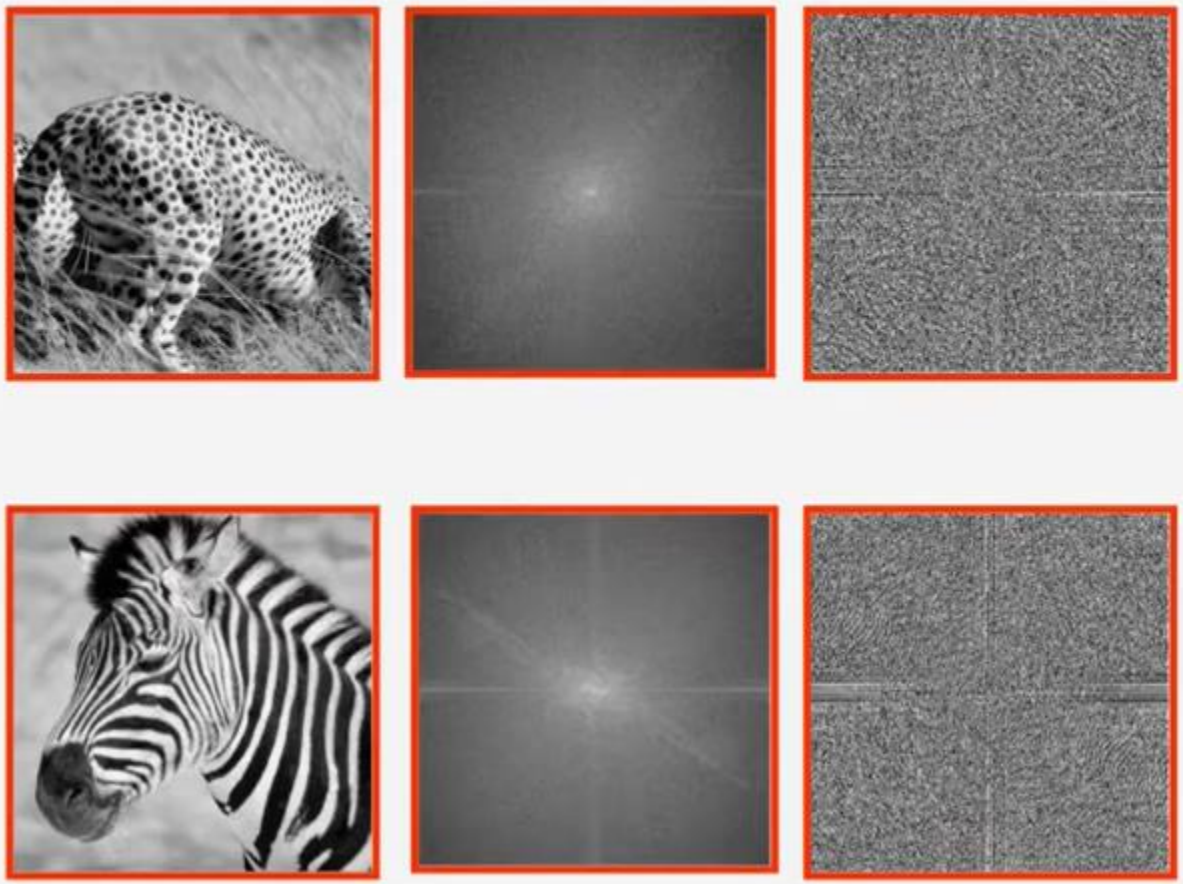
- **Low-Pass Filters:** Filters that allow low frequencies to pass through (block high frequencies). Example?
  - Gaussian filter
- **High-Pass Filters:** Filters that allow high frequencies to pass through (block low frequencies). Example?

Low-Pass and High-Pass Filters

- **Low-Pass Filters:** Filters that allow low frequencies to pass through (block high frequencies). Example?
  - Gaussian filter
- **High-Pass Filters:** Filters that allow high frequencies to pass through (block low frequencies). Example?
  - Edge filter

Which has more information: Magnitude or Phase?

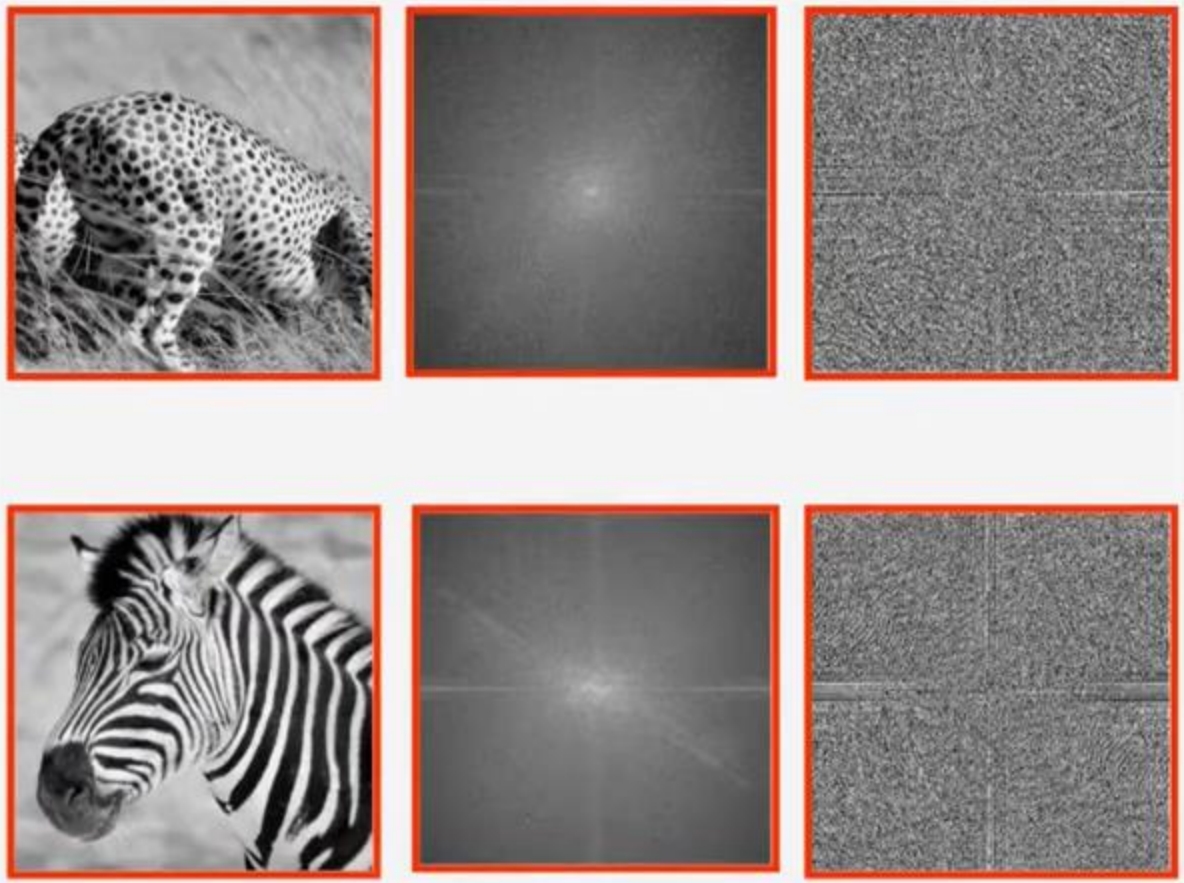
Magnitude Phase



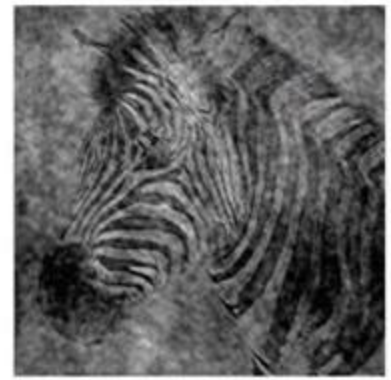
Swap phase and reconstruct?

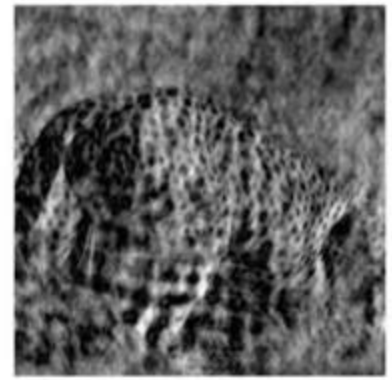
Which has more information: Magnitude or Phase?

Magnitude Phase



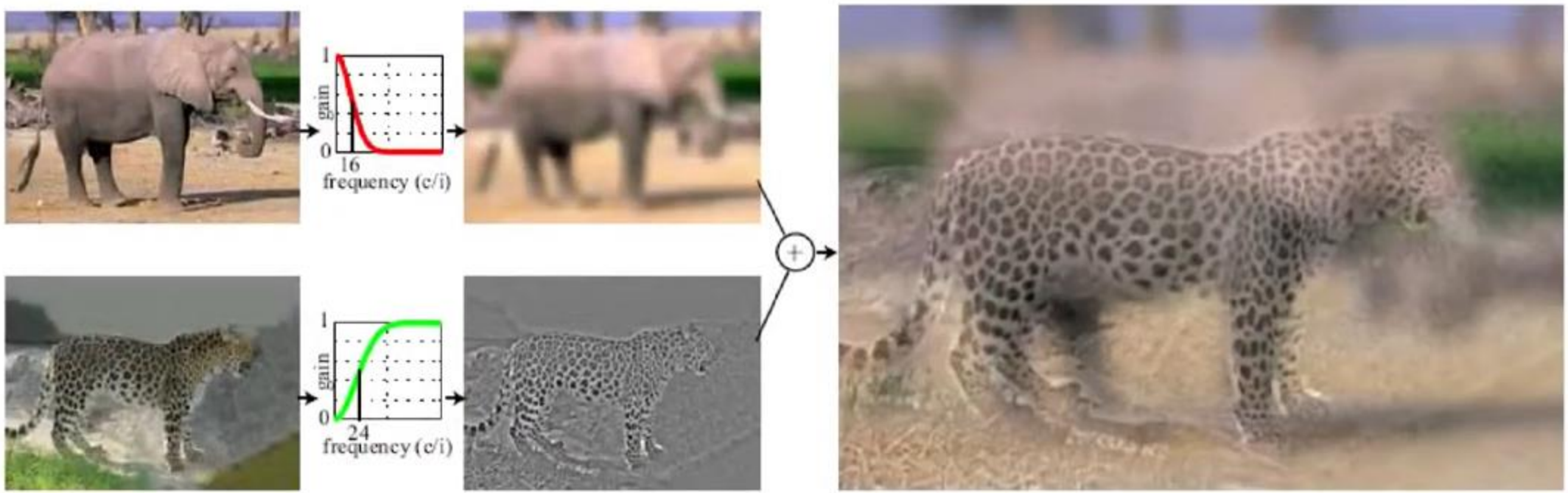
Swap phase and reconstruct?





Credit: Forsyth and Ponce, Computer Vision: A Modern Approach, 2003

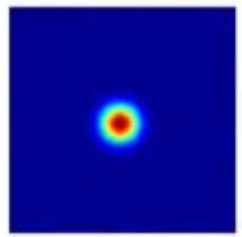
Hybrid Images



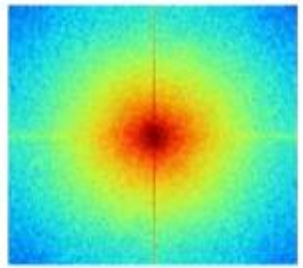
Credit: A. Oliva, A. Torralba, P.G. Schyns, "Hybrid Images," SIGGRAPH 2006

Exercise: Match spatial domain image to Fourier magnitude image

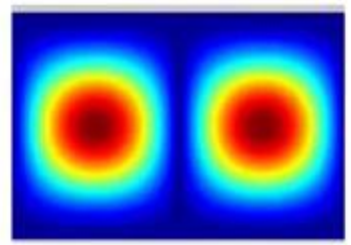
1



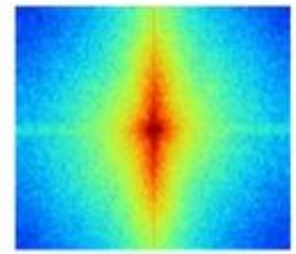
2



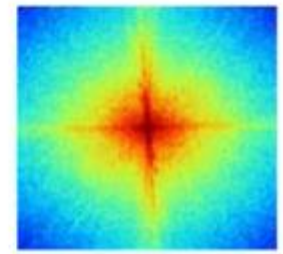
3



4



5



A



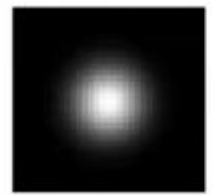
B



C



D



E



Homework

Readings

- Chapter 3.4, Szeliski, *Computer Vision: Algorithms and Applications*
- For more information on fourier transforms:
  - http://www.thefouriertransform.com/
  - http://betterexplained.com/articles/an-interactive-guide-to-the-fourier-transform/
  - http://wwwpub.zih.tu-dresden.de/~ds24/lehre/bvme_ss_2013/ip_05_fourier.pdf
- Other links provided on respective slides

Reference

- https://www.youtube.com/watch?v=MIFuBTrC6N4